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Current guidelines in hypertension management with special focus on management of extremely high blood pressure: a systematic review

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Abstract

Purpose Hypertension (HTN) represents a significant global health issue and markedly elevates the risk of cardiovascular diseases, stroke, and chronic kidney disease. This study examined various hypertension management guidelines, including the management of hypertensive crises, hypertension in adults and the elderly, and hypertension-related complications. Additionally, it examined blood pressure (BP) objectives.

Materials and methods A comprehensive search was conducted in PubMed, Scopus, and Web of Science databases, with a publication period restriction of 2010–2025. Seventeen guidelines were included and reviewed. Data on managing hypertensive crises and increased BP, including optimal BP goals, medications, and follow-up, were extracted. Data on the treatment of hypertension in patients with various End-Organ Damage (EOD) were also collected.

Results BP ≥ 180/120 mmHg with EOD has been defined as an emergency, requiring a reduction of BP by 20–25% within the first hour, followed by a gradual decrease. Ischemic stroke was the most commonly discussed EOD ($n=8$), with a target BP range of 130–180/90–105 mmHg. Elevated BP has been defined as BP ≥ 130/80 mmHg to ≥ 140/90 mmHg ($n=12$), and guidelines recommend different management strategies.

Conclusion Little data exist on how to treat high blood pressure in inpatient settings and hospitalized patients, despite the existence of sufficient guidelines and statements offering recommendations for controlling hypertension in outpatient settings. Although the guidelines suggest that patients with markedly elevated BP should be treated in inpatient settings, further research is needed to close the knowledge gaps in this area.

Keywords Hypertension, Emergency, Urgency, Asymptomatic elevated BP, Guideline

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Introduction

Hypertension (HTN) represents a significant global health issue and markedly elevates the risk of cardiovascular diseases, stroke, and chronic kidney disease (CKD) [1]. Risk factors comprise lifestyle elements such as obesity, excessive consumption of alcohol, lack of physical activity, genetic susceptibility, and age [2]. Uncontrolled HTN adversely affects multiple organs, including the heart, brain, kidneys, and blood vessels.

First-line management for asymptomatic hypertension generally includes lifestyle modifications, such as dietary alterations (e.g., the DASH diet), regular physical activity, and weight control. When lifestyle modifications are insufficient, pharmacological interventions will be initiated. Antihypertensive medications are taken orally, including thiazide diuretics, angiotensin-converting enzyme (ACEi) inhibitors, angiotensin II receptor blockers (ARBs), calcium channel blockers (CCBs), and beta-blockers (BBs). These drugs are more commonly prescribed in outpatient settings. In hypertensive emergencies, treatment typically requires hospitalization and the administration of medications through intravenous (IV) routes. Common side effects of the antihypertensive drugs vary depending on the medication class, including dizziness, fatigue, cough, and edema.

HTN continues to be a major global cause of illness and death. However, international guidelines vary in their definitions, diagnostic thresholds, and management strategies. These differences can lead to confusion in clinical practice and obstruct the consistent application of evidence-based care. This study aims to systematically compare the major HTN guidelines to identify key similarities and discrepancies. It seeks to clarify their implications for patient management and highlight opportunities for harmonizing global HTN care.

Methods

The Systematic review was carried out according to the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines and registered on PROSPERO (ID: CRD42024611815).

Data sources and search strategy

A comprehensive search was conducted in the PubMed, Scopus, and Web of Science databases, with a publication period restriction of 2010–2025. The keywords used are summarized in Table 1. Two researchers independently conducted the search. The significance level of the studies was further evaluated by examining the titles, abstracts, and full texts of the publications. A total of 17 guidelines were included and reviewed.

Table 1 Keywords used for searching the databases

Treatment terms	"Treatment" OR "Management" OR "Guidelines" OR "Therapeutics"
Population terms	"Hypertension" OR "Hypertensive patients" OR "High Blood Pressure"

Eligibility criteria

The studies were included or excluded per the defined inclusion and exclusion criteria. Guidelines or statements on the management of HTN, including treatment recommendations, were included. The articles must be in the English language and published after 2010. Only the recent version of each guideline is included.

The following exclusion criteria were followed:

(1) Case reports, narrative reviews, cohort studies, and randomized controlled trials.
(2) Articles in a language other than English.
(3) Articles on children or pregnant women.
(4) Articles on pre-operative management of HTN.
(5) Articles focusing on the management of HTN in patients with hypertensive complications exclusively.
(6) Unavailability of full texts.

Study selection

Rayyan software was used to organize the search results and remove duplicates. Eight authors independently screened 162 non-duplicated records, and any conflicts were resolved through discussion among the authors. Figure 1. Depicts the study selection process.

Data extraction

Eight authors of the research team extracted the required data: study year, study setting, management of hypertensive emergencies, urgency, increased BP, optimal BP goals, medications, and follow-up. Data on the treatment of hypertension in patients with various end-organ damage (EOD) was also extracted. The results of the included articles are discussed in Tables 2, 3, 4 and 5, and 6. The first author investigated the extracted data and settled disagreements among the other authors.

Quality assessment

The AGREE II (Appraisal of Guidelines for Research & Evaluation II) instrument [20–22] was used to assess the quality of the included guidelines. The quality analysis results ranged from 2.5 to 6. The results are given in the supplementary material.

Statistical analysis

All data were extracted onto a predesigned Microsoft Excel sheet and presented as percentages, means, and standard deviations for appropriate variables.

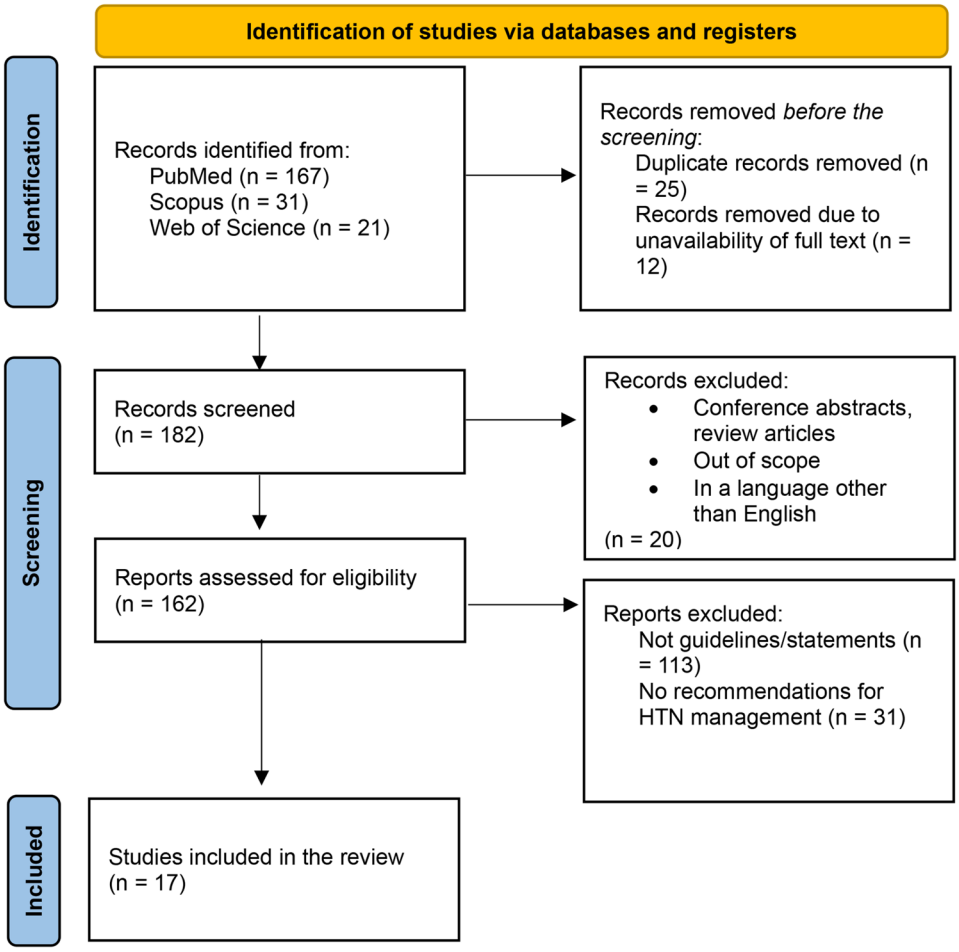


Fig. 1 Depicts the study selection process

Results

The current systematic review included 17 guidelines from multinational organizations (WHO, the United States of America, England, Australia, Canada, Japan, Korea, and Brazil) [3–18]. The guidelines included are given in Table 2.

Management of hypertensive emergencies

The management of hypertensive emergencies is discussed in eight guidelines. EOD was required in most guidelines ($n=7$). First-line medications recommended for managing hypertensive emergencies include labetalol, nicardipine, esmolol, sodium nitroprusside, clevidipine, nitroglycerin (NTG), hydralazine, metoprolol, propranolol, and phentolamine. Management was predominantly carried out in emergency settings ($n=3$), while the Brazilian guideline and the recent AHA statement recommended intensive care unit management for monitoring ($n=2$). The guidelines recommended a reduction of BP by 20–25% within the first hour, followed by gradual reductions to a target BP of 160/100–110 mmHg within two to six hours and further normalization to less than 140/90 mmHg over 24–48 h. Follow-up recommendations ranged from immediate ICU admission for intensive monitoring to discharge on oral medications and regular follow-up, emphasizing the importance of continuity of care. Additional recommendations highlighted the importance of continuous BP monitoring and of individualizing treatment based on comorbidities and specific clinical outcomes. Table 3 presents the management of hypertensive emergencies according to various guidelines.

Three guidelines defined hypertensive urgency. The threshold BP to define urgency was ≥180/120 mmHg (without EOD). The Brazilian guideline specifically mentioned the use of Clonidine for hypertensive urgencies. The guidelines recommended starting treatment within a few hours of diagnosis. Table 4 presents the management of hypertensive urgency according to the guidelines.

Management of end-organ damages

Of the 17 guidelines reviewed, nine recommended managing HTN with evening dosing. The recommended management strategies and target BP ranges varied

Table 2 Representing the included guidelines

The Management of Elevated Blood Pressure in the Acute Care Setting: A Scientific Statement from the American Heart Association	AHA 2024 [3]
2014 evidence-based guideline for the management of high blood pressure in adults: report from the panel members appointed to the Eighth Joint National Committee (JNC 8)	JNC 2014 [4]
Clinical policy: critical issues in the evaluation and management of adult patients in the emergency department with asymptomatic elevated blood pressure.	ACEP 2013 [5]
Guideline for the Diagnosis and Management of Hypertension in Adults – 2016.	Australia 2016 [6]
Investigation and management of young-onset hypertension: British and Irish Hypertension Society Position statement	BISH 2024 [7]
Screening for Hypertension in Adults: US Preventive Services Task Force Reaffirmation Recommendation Statement	USPS 2021 [8]
Diagnosis and Management of Hypertension in the Primary Care Setting Work Group, Department of Veterans Affairs, U.S. Department of Defense. VA/DoD Clinical Practice Guideline for the Diagnosis and Management of Hypertension in the Primary Care Setting. U.S. Government Printing Office; 2020.	VA 2020 [9]
Canada's 2020 Comprehensive Guidelines for the Prevention, Diagnosis, Risk Assessment, and Treatment of Hypertension in adults and children	HC 2020 [10]
Brazilian guidelines of hypertension- 2020	SBC 2020 [11]
2024 ESC Guidelines for the management of elevated blood pressure and hypertension	ESC 2024 [12]
Guideline for the Pharmacological Treatment of Hypertension in Adults. 2021.	WHO 2021 [13]
The Japanese Society of Hypertension Guidelines for the Management of Hypertension (JSH 2019)	JSH 2019 [14]
2024 European Society of Hypertension clinical practice guidelines for the management of arterial hypertension	ESH 2023 [15]
Executive summary of the 2020 clinical practice guidelines for the management of hypertension in the Philippines	Philippines 2020 [16]
National Institute for Health and Care Excellence. Hypertension in Adults: Diagnosis and Management. NICE guideline [NG136]	NICE 2019 [17]
The 2022 focused update of the 2018 Korean Hypertension Society Guidelines for the management of hypertension	KHS 2022 [18]
AHA/ACC/AANP/AAPA/ABC/ACCP/ACPM/AGS/AMA/ASPC/NMA/PCNA/SGIM Guideline for the Prevention, Detection, Evaluation, and Management of High Blood Pressure in Adults: A Report of the American College of Cardiology/American Heart Association Joint Committee on Clinical Practice Guidelines	ACC/AHA [19]

depending on the specific type of EOD. Ischemic stroke was the most commonly discussed condition ($n=9$), with a target BP range of 130–180/90–105 mmHg. Some articles also recommended maintaining BP < 185/110 mmHg before fibrinolysis, then < 180/105 mmHg, aiming to reduce BP by 15% during the first 24 h following stroke onset. Other EODs included in the guidelines were intracranial hemorrhage ($n=7$), with a target BP range of < 140–220/80 mmHg; adrenergic crisis ($n=5$), with a target SBP < 140 mmHg; acute coronary syndrome ($n=3$), with a target BP range of < 130–140/80 mmHg and; acute pulmonary edema ($n=3$), with a target SBP < 140 mmHg; malignant HTN ($n=3$), with a target reduction in mean arterial pressure (MAP) by 20–25% within 1 h and DBP reduction to 100–110 mmHg within 24 h; aortic dissection ($n=3$), with a target BP range of SBP 100–120 mmHg; and hypertensive encephalopathy ($n=3$), with a target MAP reduction of 20–25% within 1–3 h. These findings demonstrate that each type of end-organ damage requires distinct BP management strategies. Table 5 discusses the management of different EODs.

Management of asymptomatic HTN

Elevated BP has been defined in 13 guidelines, ranging from ≥ 130/80 mmHg to ≥ 140/90 mmHg. However, the ACEP guideline classified BP > 160/100 as elevated BP. The guidelines recommend starting treatment when BP is ≥ 140/90 mmHg ($n=4$). The 2025 ACC/AHA guideline recommended starting treatment if the BP remains > 130/80 mmHg following lifestyle modifications for 3–6 months. The guideline also recommends against prescribing additional oral or intravenous antihypertensives in inpatient settings when the BP is > 180/120 mmHg. However, the AHA 2024 statement and the Brazilian guideline recommend initiating treatment if BP exceeds 180/100 mmHg. ACEi, ARBs, CCBs, thiazide diuretics, potassium-sparing diuretics, and BBs are the first-line drugs used. The AHA 2024 statement recommended using labetalol, nicardipine, clevidipine, fenoldopam, NTG, nitroprusside, esmolol, and hydralazine as the first-line drugs. The follow-up period for asymptomatic elevated BP varied across guidelines, ranging from 3 months to 1 year. The BISH and Philippines guidelines recommended a follow-up period of 3 months. The KSH, HC, WHO, and JSH guidelines recommend a monthly follow-up immediately after initiating treatment, followed by follow-ups every 3 months to a year. The JSH guideline recommends a follow-up period of 1 to 3 months, depending on the patient's risk. The ESC guideline advised immediate treatment upon diagnosis, with monthly follow-ups until BP stabilization, then yearly follow-ups. Table 6 represents the management of elevated BP.

Discussion

In this systematic review, most included guidelines specified elevated BP as ≥ 130–140 mmHg SBP or ≥ 80–90 mmHg DBP. Most of the guidelines discussed

Table 3 Representing the management of hypertensive emergencies as per different guidelines

Guidelines	BP threshold	EOD	Drugs	Route	Site	Treatment	Follow up
AHA 2024	180/120 mmHg	Yes	Labetalol, nicardipine, sodium nitroprusside, clevidipine, esmolol, NTG	IV	Emergency or inpatient critical care	Immediate initiation, aiming to lower MAP by 20–25% within the first hour	Continuous monitoring in the ICU; oral medications upon stabilization and discharge planning
Bish 2024	180/120 mmHg	Yes	N/A	N/A	Outpatient	N/A	3 months
HC 2020	DBP ≥ 130 mmHg	Yes	N/A	N/A	N/A	N/A	N/A
SBC 2020	180/120 mmHg	yes	Sodium nitroprusside, NTG, metoprolol	IV	emergency	Reduce BP by 25% in 1 st hr, followed by 160/100–110 mmHg at 2–6 h and return to normal values in 24–48 h	Preferential ICU admission
ESC 2024	180/110 mmHg	Yes	Nitroprusside, esmolol, clevidipine	IV	emergency	Immediate treatment	N/A
JSH 2019	180/120 mmHg	Yes	Vasodilators: nicardipine, diltiazem, NTG, sodium nitroprusside, hydralazine. Sympatholytic phentolamine, propranolol	IV	ICU facilities with specialists	Immediate initiation. A decrease in BP of 25% during the first 1 h and to a level of 160/100 mmHg within the next 2–6 h. Subsequently, BP should be carefully decreased to < 140/90 mmHg over 24–48 h	N/A
NICE 2019	180/120 mmHg	Yes	N/A	N/A	N/A	N/A	N/A
ACC/AHA 2025	180/120 mmHg	No	No intervention recommended	N/A	N/A	N/A	N/A

Abbreviations: *DBP* Diastolic Blood Pressure, *ICU* Intensive Care Unit, *IV* Intravenous, *MAP* Mean Arterial Pressure, *NTG* Nitroglycerin, *SBP* Systolic Blood Pressure

Table 4 Representing the management of hypertensive urgencies as per different guidelines

Guidelines	Threshold	Drugs
BISH 2024	> 180/120 mmHg	N/A
SBC 2020	≥ 180/120 mmHg	Captopril, Clonidine
NICE 2019	≥ 180/120 mmHg	N/A

the management of elevated BP in outpatient settings. However, according to the recent AHA statement, asymptomatic elevated BP without EOD is treated non-pharmacologically. Regular follow-up is advised to monitor for EOD. Elevated BP is also encountered in the inpatient setting. However, most guidelines did not mention the recommendations for managing inpatient elevated BP. Various factors, most importantly anxiety and pain, along with acute medical conditions, fluid overload, and medications, can contribute to elevated inpatient BP. Adherence to the medication regimen is also an important consideration because patients who have trouble with outpatient medication adherence are more likely to have high BP during hospitalization [19].

According to the AHA 2024 statement on managing high BP in an acute care setting, elevated inpatient BP is defined as ≥ 130 mmHg SBP or ≥ 80 mmHg DBP, consistent with the 2025 HTN clinical practice guidelines definition of HTN. Objective terminology of elevated BP has been proposed, as the use of subjective emotive language, such as hypertensive crisis and urgency, led to unnecessary antihypertensive treatment. Elevated inpatient BP (≥ 130 mmHg SBP or ≥ 80 mmHg DBP), which is not markedly elevated (SBP ≤ 180 mmHg and DBP ≤ 110 mmHg), is called asymptomatic elevated BP. While markedly elevated inpatient BP, that is ≥ 180 mmHg SBP or ≥ 110–120 mmHg DBP (formerly hypertensive crisis), can be asymptomatic (formerly hypertensive urgency) if there is no new or worsening EOD, on the other hand, it is called hypertensive emergency if there is new or worsening EOD. However, EOD can manifest even below the 180/110–120 mmHg threshold [3, 23].

In a recent retrospective study of 133, 760 veterans using data from a Veterans Administration (VA) hospital, it was found that pro re nata (PRN) BP medication during a hospital stay for asymptomatic elevated BP was associated with significantly increased risk of acute kidney injury (AKI), especially in those who received IV compared to oral or both oral and IV medication. PRN medication also led to a 1.5-fold greater risk of greater BP reduction, and composite outcome of myocardial infarction, death, and stroke [24]. In a target trial emulation study by Anderson et al., intensive anti-hypertensive treatment strategies among adults ≥ 65 years of age admitted for non-cardiac reasons, intensive BP treatment

Table 5 Representing the management of end organ damage as per different guidelines

Guidelines	End-organ Damage	Target BP	Treatment
AHA 2024	• Stroke, Hypertensive encephalopathy, and intracranial hemorrhage • Acute aortic dissection, Preeclampsia, HELLP, eclampsia • Grade III-IV Keith-Wagener-Barker hypertensive retinopathy • AKI, Thrombotic microangiopathy	• 130 < SBP < 180 mmHg • 120 < SBP < 160 mmHg DBP < 105 mmHg • SBP < 180 mmHg • N/A	• Labetalol, nicardipine • Esmolol/nitroprusside/NTG/nicardipine/labetalol/magnesium sulfate/hydralazine • N/A • Labetalol, nicardipine, clevidipine, and fenoldopam
BISH 2024	• Aortic dissection, or post-stroke	• As per NICE 2023 guidelines	• N/A
SBC 2020	• Ischemic stroke • Intracranial hemorrhage • ACS • Acute aortic dissection • APE • Malignant HTN • Hypertensive encephalopathy • Adrenergic crisis	• < 185/110 mmHg recommended before fibrinolytic therapy • SBP < 180 mmHg • 120 < SBP < 130 mmHg, DBP 70–80 mmHg • 100 < SBP < 120 mmHg • N/A • N/A • N/A • N/A	• N/A • N/A • BBs, NTG • N/A • NTG, nitroprusside loop diuretics • Diuretics, renin-angiotensin system blockers, BBs, hydralazine, clonidine, methyldopa, and CCBs • Nitroprusside • Alpha1-blockers, BBs, CCBs, ACEIs, and central action agonists
ESC 2024	• Ischemic stroke • Intracranial hemorrhage • Adrenergic crisis	• BP should be lowered to < 185/110 mmHg before thrombolysis and then maintained at < 180/105 mmHg over the following 24 h • SBP < 180 mmHg • N/A	• N/A • N/A • Alpha-1-blocker
JSH 2019	• Ischemic stroke • Intracranial hemorrhage • Hypertensive encephalopathy • ACS • APE • Malignant HTN • Aortic dissection • Adrenergic crisis	• Hyperacute phase - <180/105 mmHg/85–90% of pre-treatment value. Acute phase- 85% of pre-treatment value • 130/80 mmHg • A ~ 25% decrease in BP over the first 2–3 h • SBP < 140 mmHg • 10–15% decrease in SBP • DBP of 100–110 mmHg • SBP of 100–120 mmHg • SBP < 140 mmHg within 1 h after treatment initiation	• Nicardipine, diltiazem, NTG, or nitroprusside • CCBs, ACEi, ARBs or diuretics • Nicardipine, diltiazem, nitroprusside • Nitrites, BBs, ACEi • Nitrates, carperitide/nicorandil, Furosemide • ACEi, ARBs • CCBs/NTG/nitroprusside in combination with a BB • Phentolamine, doxazosin
HC 2020	• Ischemic stroke • Adrenergic crisis/pheochromocytoma	• < 140/90 mmHg • N/A	• ACEi, and thiazide/thiazide like diuretics/combination therapy • MR antagonists, surgery
Philippines	• Ischemic stroke • Intracranial hemorrhage	• Acute Ischemic Stroke, eligible for IV thrombolysis- <185/110 mmHg before thrombolysis and 180/105 mmHg in the next 24 h. Not eligible for IV thrombolysis or mechanical thrombectomy- reduce the BP by 15% during the first 24 h • SBP < 180 mmHg.	• Nicardipine, labetalol • Nicardipine, labetalol

Table 5 (continued)

Guidelines	End-organ Damage	Target BP	Treatment
KHS	• Ischemic stroke • Intracranial hemorrhage	• Patients treated with thrombolysis < 185/100 mmHg before treatment and maintain BP < 180/105 mmHg during the first 24 h. Patients undergoing endovascular recanalization- <180/105 mmHg. Patients not eligible for any- BP lowering by approximately 15% during the first 24 h. • Patients presenting < 6 h of onset- rapid lowering of SBP to 140 mmHg.	• Thiazide diuretics, ACEi, ARBs, or combination therapy • N/A
ACC/AHA 2025	• Intracerebral hemorrhage • Ischemic stroke	• For adult patients with acute spontaneous ICH who present with SBP between 150 and 220 mm Hg, it can be beneficial to immediately lower SBP to 130 to < 140 mm Hg for at least 7 days after ICH to improve functional outcomes, but stop antihypertensive medications if SBP < 130 mm Hg. • In patients undergoing successful brain reperfusion with endovascular treatment for a large vessel occlusion, lowering SBP < 140 mm Hg within the first 24 to 72 h after reperfusion can worsen long-term functional outcome.	• N/A • N/A

Abbreviations: *ACEi* Angiotensin Converting Enzyme Inhibitor, *ACS* Acute Coronary Syndrome, *AKI* Acute Kidney Injury, *APE* Acute Pulmonary Edema, *ARBs* Angiotensin Receptor Blockers, *BBs* Beta-blockers, *CCBs* Calcium Channel Blockers, *DBP* Diastolic Blood Pressure, *HELLP* Hemolysis Elevated Liver Enzyme Levels, and Low Platelet Levels, *NTG* Nitroglycerin, *SBP* Systolic Blood Pressure

increased the risk of a composite outcome of mortality, AKI, stroke, troponin elevation, BNP elevation, and transfer to the intensive care unit [25]. However, benefits may outweigh the risk for treatment of certain groups with asymptomatic elevated inpatient BP, especially in those with persistent markedly elevated inpatient BP readings, SBP ≥ 180 and DBP ≥ 110–120 mmHg, who have high outpatient BPs, as well as those at high risk of cardiovascular disease, initiating or intensifying antihypertensive and close out-patient follow up as typical inpatient stay is unlikely to be sufficient to fully assess the effect of a single medication change [3].

In the review, most HTN guidelines primarily focus on the outpatient setting. The SBC 2020 guidelines had a wider breadth, covering outpatient, ambulatory, and emergency-based management, whereas the AHA 2024 [6] and ACEP 2013 guidelines specifically addressed HTN management in acute care settings. The AHA 2024 statement emphasizes that while hypertensive emergencies require immediate treatment, the benefit of initiating antihypertensive medications for asymptomatic elevated inpatient BP remains unclear. However, HTN guidelines for emergency and inpatient settings frequently include a higher risk of overtreatment. The potential risks associated with overtreatment have been particularly highlighted in patients aged 65 years or older [26]. Guidelines for managing HTN in emergency and inpatient settings also emphasize the importance of continuous BP monitoring and adequate patient follow-up to improve treatment outcomes. According to the 2025 ACC/AHA HTN Guideline, adults starting or changing antihypertensive medication should undergo follow-up examinations monthly until BP control is achieved [19]. This review suggests ICU monitoring or outpatient care within 3 months, whereas a study by Thomas et al. recommends outpatient follow-up for non-admitted patients without specifying a timeline. This difference highlights the variation in long-term management recommendations and the need for further standardization across clinical guidelines [25, 27].

HTN-related EODs are a more sensitive measure of disease onset, progression, and prognosis. Hypertensive emergencies are managed based on the nature of presentation and EOD. Guidelines suggest that markedly elevated BP without acute HTN-mediated organ damage can usually be treated with oral BP-lowering medication or adjustment of their current BP-lowering medication [28]. Oral medications such as captopril, labetalol, and nifedipine can be used to achieve a timely, gradual reduction in BP.

Emergent situations such as hemorrhagic stroke, acute intracerebral hemorrhage (ICH), require more aggressive and tailored management without bringing down the BP too quickly. The ACC/AHA 2025 guideline emphasized BP management in these situations. The guideline suggests lowering the BP to 130–140 mmHg in patients with ICH within the first 7 days, but stopping the treatment if it falls below 130 mmHg [19]. Similarly, in patients with ischemic stroke undergoing reperfusion therapy, lowering the BP below 140 mmHg is associated with adverse outcomes. Oral agents may be preferred in some settings, but malignant or resistant cases will require IV agents such as labetalol or nicardipine, nitroprusside, and NTG. BP-lowering targets are to be achieved within normal limits over 24 to 48 h, with graduation by the hour and minute. Acute MI, Acute aortic dissection, and hypertensive encephalopathy can be managed with IV agents - labetalol, esmolol, NTG, and fenoldopam in combinations. Labetalol and nicardipine, along with fenoldopam and NTG, are primarily used in combination for treating

Table 6 Representing the management of elevated BP as per different guidelines

Guidelines	Range	Threshold	Drugs	Follow up	Inpatient management
AHA 2024	≥ 130/80 mmHg	180/100 mmHg	Labetalol, nicardipine, clevidipine, fenoldopam, NTG, nitroprusside, esmolol, hydralazine.	Close monitoring in the hospital setting, followed by transition to oral antihypertensives post-discharge, and outpatient follow-up.	N/A
ACEP 2013	≥ 160/100 mmHg	N/A	N/A	N/A	N/A
Australia 2016	≥ 140/90 mmHg	N/A	N/A	Home or ambulatory monitoring.	N/A
BISH 2024	≥ 135/85 mmHg	160/100 mmHg	ACEi, ARBs, CCBs, thiazides	3 months	N/A
USPS 2021	≥ 130–140/80-90mmHg	N/A	ARBs	Regular yearly screening of adults over 40 years old and high-risk groups. Screening every 3–5 years for patients aged between 18 to 39, with no comorbidities	N/A
VA 2020	≥ 130/90 mmHg	130/90 mmHg	N/A	Periodic follow-up	N/A
HC 2020	≥ 130/80 mmHg	(i) The mean OBPM is 140/90 mmHg in patients with macrovascular EOD, DM, or CKD (ii) At visit 3, the mean OBPM is 160/100 mmHg (iii) At visit 4 or 5, the mean OBPM is 140/90 mmHg	Thiazides, BBs, ACEi, ARBs, or long-acting CCBs	New onset HTN- 1–2 months HTN with target BP reached 3–6 months	N/A
ESC 2024	Ambulatory BP: 120–134/70–84 mmHg OBP: 120–139/70–89 mmHg	140/90 mmHg	Low-dose double combination therapy, ACEi, ARBs, CCBs, diuretics (Class I)	Monthly follow-up till stabilization of BP, followed by a yearly follow-up.	N/A
ESH 2023	130–139/80–89 mmHg	N/A	ACEi/ARBs + CCBs or/+ thiazide diuretic. BBs as monotherapy or at any step of combination therapy	Follow up 3 months after treatment initiation. Short follow-up of 3 months-1 year to establish optimal BP control. Long-term follow-up of > 1 year to maintain optimal BP control.	N/A
Philippians 2020	≥ 140/90 mmHg	140/90 mmHg	ACEi or ARBs, CCBs, thiazide/thiazide-like diuretics	3 months	N/A
NICE 2019	135–140/85–90 mmHg	N/A	ACEi, ARBs, CCBs, thiazide diuretics	N/A	N/A
KHS 2022	≥ 140/90 mmHg	N/A	ACEi, ARBs	New onset HTN- 1–2 months HTN with target BP reached- 3–6 months.	N/A
ACC/AHA 2025	> 130/80 mmHg	N/A	N/A	N/A	N/A

Abbreviations: *ACEi* Angiotensin Converting Enzyme Inhibitor, *ARBs* Angiotensin Receptor Blockers, *BBs* Beta-blockers, *CCBs* Calcium Channel Blockers, *CKD* Chronic Kidney Disease, *DM* Diabetes Mellitus, *NTG* Nitroglycerin, *OBPM* Office Blood Pressure Measurement

congestive heart failure, Eclampsia, Pheochromocytoma, Acute pulmonary edema, Acute ischemic stroke/intracerebral bleed, and Acute renal failure/MAHA [19, 29].

However, there are exceptional cases where conventional treatment doesn't work. Pickering syndrome, characterized by renal artery stenosis, worsens after treatment with ACE-Is and ARBs [30]. Similarly, aggressive BP lowering is associated with dysregulation of cerebral blood flow, and the use of sublingual nifedipine has been known to cause serious adverse effects [31, 32]. It is also essential to consider the cerebral blood flow autoregulation curve when administering antihypertensive therapy in older adults, especially in emergency or urgent care settings, as highlighted by Buonacera et al. [32]. Spironolactone is used in resistant HTN to manage volume overload. However, it is associated with hyperkalemia, particularly in combination with ACE inhibitors or ARBs in older individuals and patients with CKD. This requires careful monitoring of potassium levels. This pattern occurs more frequently in elderly patients, patients with chronic renal failure, and Pickering syndrome [33]. In cases of resistant hypertension, particularly among younger individuals, a thorough evaluation to exclude even rare secondary forms of hypertension is mandatory

to guide appropriate management [34]. Similarly, the use of RAAS inhibitors, in combination therapy, prevents cardiac remodeling, thereby reducing the risk of atrial fibrillation and the occurrence of ischemic stroke [35].

Most HTN guidelines are based on similar evidence; however, variations exist due to regional differences in epidemiology, healthcare infrastructure, and resource availability. The 2025 ACC/AHA guideline adopts lower thresholds to promote early intervention and reduce cardiovascular risk in high-resource settings [19]. In contrast, the ESC/ESH guideline focuses on individualized management to avoid overtreatment in lower-risk populations [12, 15]. Meanwhile, the WHO 2021 guideline emphasizes feasibility and cost-effectiveness for low- and middle-income regions, where access to medications and healthcare services is often limited [13]. These differences reflect contextual adaptations rather than scientific disagreement, underscoring the need for hypertension management strategies that are coherent on a global scale yet adaptable to local circumstances.

Healthcare professionals generally adhere to the guidelines established in their own country or region, which can limit the direct clinical applicability of a simple compilation of multiple HTN guidelines. Therefore, this review is intended not only as a reference for clinicians but also to inform policymakers, guideline developers, and educators about global trends and variations in hypertension management. By highlighting similarities and differences among guidelines, this work can support efforts to harmonize policies, adapt evidence-based recommendations to local contexts, and develop educational resources for both trainees and practicing clinicians [3–19].

Conclusion

There are sufficient guidelines and statements offering recommendations for controlling hypertension in outpatient settings and managing hypertensive crises. Although the AHA's 2024 statement suggests that patients with markedly elevated BP should be treated in inpatient settings, more research is needed to close the knowledge gaps in this area. The absence of standardized inpatient protocols creates variability in clinical practice, often leading to overtreatment or delayed intervention. Therefore, prospective studies are needed to establish evidence-based strategies for inpatient BP management, optimize patient outcomes, and avoid potential harm from inappropriate treatment approaches.

Abbreviations

DBP	Diastolic Blood Pressure
ICU	Intensive Care Unit
IV	Intravenous
MAP	Mean Arterial Pressure
NTG	Nitroglycerin
SBP	Systolic Blood Pressure

Supplementary Information

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Supplementary Material 1.

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Authors' contributions

R.V. conceptualized the study and performed the data analysis. S.S.M.A., P.A., S.T., S.K.A., D.A., Y.T., R.P.R., R.G.B., and N.D. contributed to data collection. All authors contributed to writing the manuscript. G.K. reviewed and approved the final version of the manuscript. R.G.B. served as the corresponding author.

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Data availability

All data generated or analysed during this study are included in this published article [and its supplementary information files].

Declarations

Ethics approval and consent to participate

Not applicable in our institution.

Consent for publication

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Competing interests

The authors declare no competing interests.

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